SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

IMPLEMENTATION PLAN & REPORT

Mwange Musa Muthami

Registration Number: 23/05037

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Supervisor: Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

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1. IMPLEMENTATION OVERVIEW

This document describes the implementation approach, technologies used, modules developed, challenges encountered, and deployment details for the Kyamatu Primary School Management System.

1.1 Technology Stack

Layer	Technology	Version
Frontend	React	18.3
Build Tool	Vite	7.3
Styling	Tailwind CSS	3.4
State Management	Zustand	5.0
Backend	Express.js	4.21
Runtime	Node.js	20+
Database	PostgreSQL	15
ORM	Prisma	5.22
Authentication	JWT + bcrypt	
Hosting (FE)	Cloudflare Pages	
Hosting (BE)	Render	
Hosting (DB)	Supabase	

2. MODULES IMPLEMENTED

2.1 Authentication & RBAC

JWT-based auth with access (15min) and refresh tokens (7 days). bcrypt password hashing. 6 roles: Super Admin, Admin, Teacher, Bursar, Student, Parent. RBAC middleware enforcing per-route permissions.

2.2 Student Management

CRUD operations with admission number auto-generation. Admission workflow (Pending → Approved/Rejected). Guardian linking via junction table. Search and filtering by name, class, grade.

2.3 Staff Management

CRUD for staff profiles with employee number generation. Teacher-to-class-subject assignment. Class teacher designation.

2.4 Academic Structure

Academic years with current year marking. Three terms per year. CBC grades PP1 through Grade 9. Streams and auto class generation. Subject management with unique codes.

2.5 Attendance

Bulk daily marking per class. 4 statuses: Present, Absent, Late, Excused. Unique constraint per student per date. Rate calculation and reports.

2.6 Assessment & Grading

Assessment types: CAT, Mid-term, End-term. Bulk score entry. Auto CBC competency ratings (EE/ME/AE/BE). Traditional letter grades (A-E). Weighted scoring.

2.7 Report Cards

Automated generation per student per term. Average, total, and class rank computation. Teacher and principal comments.

2.8 Fees Management

Fee structure per grade per term. Invoice generation with unique numbers. Payment recording (Cash, M-Pesa, Bank Transfer). Auto balance calculation.

2.9 Communication

Announcements with role-based targeting. Direct messaging. Read status tracking.

2.10 Timetable

Slot configuration (day, time). Subject/teacher assignment. Weekly grid display.

2.11 Dashboards

Admin: enrollment stats, attendance trends, fee summaries. Student: personal attendance, fees, timetable, social feed.

3. CHALLENGES & SOLUTIONS

Challenge	Solution
Supabase connection failures	Configured connection pooling and retry logic in Prisma
Data disappearing after refresh	Removed auto-seeding from login; seed runs only during setup
Assessments not grouped by term	Refactored query to include term relations, grouped in frontend
Dashboard showing placeholder data	Replaced hardcoded values with live API calls

4. DEPLOYMENT

Backend: Deployed on Render with auto-deploy from GitHub main branch.

Frontend: Deployed on Cloudflare Pages with auto-deploy from GitHub.

Database: PostgreSQL hosted on Supabase with managed backups.

Docker Compose available for local development environment.

5. SOURCE CODE REPOSITORY

GitHub: https://github.com/MuthamiM/KYAMATU-SMS